

Alireza Laalmanesh

Waterloo, ON · me@alirezalab.com · alirezalab.com · [LinkedIn](#) · [GitHub](#)

EDUCATION

University of Waterloo <i>MMath, Computational Mathematics (Co-op)</i>	Waterloo, Canada <i>Sep 2025 – Present</i>
• i-Comp-Data Fellowship. Advisors: Dr. David Del Rey Fernández and Dr. Leia Minaker. • Research: numerical analysis, deep learning for urban data, discrete-event simulation.	
Amirkabir University of Technology (Tehran Polytechnic) <i>B.Sc., Industrial Engineering & Information Systems</i>	Tehran, Iran <i>2019 – 2024</i>
• GPA: 3.5/4.0. Thesis: <i>Design of a Model for the Operating Room Scheduling Problem with Endogenous Uncertainty.</i>	

EXPERIENCE

Polytechnique Montréal <i>Co-op Research Assistant</i>	Montréal, Canada <i>Jun 2026 – Sep 2026</i>
• Applied optimization and ML to EMS dispatching; ran large-scale simulations of RL dispatch policies on real EMS data.	
University of Waterloo <i>Graduate Teaching Assistant, Faculty of Mathematics</i>	Waterloo, Canada <i>Sep 2025 – Present</i>
• MATH 128, MATH 106, MATH 135, MATH 115 — tutorials, office hours, and grading.	
Amirkabir University of Technology <i>Teaching Assistant</i>	Tehran, Iran <i>Sep 2023 – Jun 2025</i>
• Probability Theory, Optimization, and Systems Analysis; designed mathematical modeling assignments.	
TAV Institute (Tavenergyco) <i>Layout Design Intern</i>	Tehran, Iran <i>2024</i>
• Factory layout analysis, process-flow modeling, and material-handling capacity analysis.	
Independent / Kanoon Farhangi Amoozesh <i>Mathematics & Physics Tutor; Academic Advisor</i>	Tehran, Iran <i>Dec 2019 – Sep 2025</i>
• Tutored grades 7–12; advised students on academic planning and study habits.	

SELECTED PROJECTS

CrashVision-DL — Traffic Severity Prediction	<i>PyTorch, DeepInsight</i>
• Tabular-to-image pipeline on 7.7M records (t-SNE); Soft F-Beta loss for fatal-crash recall.	
Academic Hub Automation	<i>Docker, Linux, Microsoft 365</i>
• Local bridge from Linux research environments to Microsoft 365 via email transport.	
Scientific Content Pipeline	<i>Python, Gemini Vision, AsyncIO</i>
• ETL pipeline (PyMuPDF + Gemini Vision) converting scientific PDFs into audio notes.	
Android Hardware Bridge	<i>Linux, ADB, USB/IP</i>
• USB/IP setup turning Android devices into network-exposed hardware servers.	
Server Config Utility & CodeBase Aggregator	<i>Bash, Python</i>
• Headless server init (Tailscale, XRDP) and repo aggregation for LLM context datasets.	

TECHNICAL SKILLS

Languages	Python, C/C++, Bash
ML / AI	PyTorch, scikit-learn, XGBoost, EfficientNet, DeepInsight
Math	Numerical optimization, stochastic modeling, discrete-event simulation
Systems	Linux, Docker, Bash automation, networking protocols, AsyncIO

HONORS & AWARDS

• i-Comp-Data Fellowship; Top 1% National University Entrance Exam (~150,000 candidates); Merit Scholarship, Amirkabir University of Technology.	
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